Core Components

Phoenix includes a (fairly large) module that creates a simple component system for you. You don't have to use this module, but it has several useful components inside of it.

Open up the PhoneAppWeb.CoreComponents module at lib/phone_app_web/components/core_components.ex and check it out. This module may be a bit intimidating if you're new to web development—don't worry too much about it in that case.

The message_form component uses the input component defined in CoreComponents. This component renders the styled input—handling different input types properly—and includes an error section when the form has invalid data.

We won't go over this file too much, but it's important to know that it's here. You can change this file to fit your application, or you can use it as a pattern for your own application components.

Before we wrap this chapter, let's go over a few aspects of Phoenix that didn't make it into this book's project.

Phoenix Is More Than Controllers

Phoenix isn't an all-in-one web framework, but it does more than just HTTP requests. This chapter focuses on controllers, views, and components because those are the foundations, but there's even more that you can do with Phoenix.

We'll briefly touch on a few different major features that Phoenix offers: authentication, Phoenix Channels, and LiveView.

Authentication Generator

There's a good chance your application will need to support registering users, logging in, logging out, and restricting access to routes. Phoenix solves this by offering a generator[26] that adds authentication code to your application.

All you have to do is type mix phx.gen.auth Accounts User users to create the user system inside of the Accounts context. This will create all of the schemas, migrations, mailers, controllers, and tests. (If you run this in the book project, make sure to back up the project first.)

The Phoenix generator creates an authentication system in your app, but there are other libraries that are more like Devise—Ruby's all-in-one authentication

26. https://hexdocs.pm/phoenix/mix_phx_gen_auth.html